LJB

Reg. No. | | | | | | | | | | | [18BS2MA01]

SECOND SEMESTER B.Tech EXAMINATION, JUNE–2019
MATHEMATICS-2

Time:3 Hours **Maximum Marks:70**

Instructions:

i. Answer any FIVE full Questions. Questions 1 and 2 are compulsory.

ii. Missing data may be suitably assumed.

1.

a. Change the order of the integration and hence evaluate $\int_0^1 \int_{\sqrt{y}}^1 dx\,dy$ **04**

b. Find by double integration the area enclosed by the curve $r = a(1+\cos\theta)$ between $\theta = 0$ and $\theta = \pi$. **05**

c. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\,dz\,dy\,dx$. **05**

2.

a. Verify Green's theorem in a plane for $\oint_C \left(xy+y^2\right)dx + x^2 dy$, where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. **07**

b. Evaluate using Gauss Divergence Theorem for the vector $\vec{F} = \left(x^2 - yz\right)\hat{i} + \left(y^2 - xz\right)\hat{j} + \left(z^2 - xy\right)\hat{k}$ taken over the rectangular parallelepiped $0 \le x \le a, 0 \le y \le b, 0 \le z \le c$. **07**

3.

a. Find the Laplace transform of the function $\frac{1-e^{-at}}{t}$ **04**

b. Given $f(t) = \begin{cases} E, & 0 < t < \frac{a}{2} \\ -E, & \frac{a}{2} < t < a \end{cases}$ where $f(t+a) = f(t)$. show that **05**

$$L[f(t)] = \frac{E}{s}\tanh\left(\frac{as}{4}\right)$$

c. If $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$, express $f(t)$ in terms of unit step function and hence find its Laplace transform. **05**

OR

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4. a. Find the inverse Laplace transform of the function $\log\left(\sqrt{\frac{(s^2+1)}{(s^2+4)}}\right)$ **04**

b. Find the inverse Laplace transform of $\frac{1}{(s+1)(s^2+1)}$ by using convolution theorem. **05**

c. Solve the differential equation $y''-3y'+2y=e^{3t}$, given $y(0)=0$ and $y'(0)=0$ using Laplace transform. **05**

5. a. State and prove Cauchy-Riemann equations in Polar form. **04**

b. Show that the function $u=e^x\cos y+xy$ is harmonic and hence find its corresponding analytic function. **05**

c. If $f(z)$ is analytic function of z then show that **05**

$$\left[\frac{\partial}{\partial x}|f(z)|\right]^2+\left[\frac{\partial}{\partial y}|f(z)|\right]^2=|f'(z)|^2$$

OR

6. a. Discuss the Transformation $w=e^z$ with the help of a neat diagram. **07**

b. Find the bilinear transformation which maps the points $z=1,i,-1$ into $w=i,0,-i$. **07**

7. a. Evaluate $\int_C z^2dz$, along the curve made up of two line segments, one from $z=0$ to $z=3$ and another from $z=3$ to $z=3+i$. **04**

b. Verify Cauchy's theorem for $f(z)=ze^z$ over unit circle with origin as center. **05**

c. Evaluate $\int_C \frac{e^{2z}}{(z+1)^2(z-2)}dz$ where C is the circle $|z|=3$. **05**

OR

8. a. Find poles and residues of the function $f(z)=\frac{z}{(z+1)(z-2)^2}$. **04**

b. Expand $f(z)=\frac{z+1}{(z+2)(z+3)}$ as a Laurent's series valid for $2<|z|<3$. **05**

c. Using Cauchy's residue theorem evaluate $\int_C \frac{\sin \pi z^2+\cos \pi z^2}{(z-1)^2(z-2)}dz$ where C is the circle $|z|=3$. **05**